

“Supply Chain Data Analysis”



Python + MySQL + Power Bi + Excel Project

Supply Chain Data Analysis

Project Title: “Supply Chain Performance Analysis: Strategic Insights for Business Growth”

2. Project Background: This project focuses on performing a comprehensive data-driven analysis of a real-world supply chain dataset containing **1,80,519** orders spanning multiple markets, regions, and customer segments across the globe. By leveraging advanced data analytics techniques, this study aims to uncover hidden patterns, inefficiencies, and growth opportunities within the supply chain ecosystem.

Dataset Summary	
Detail	Info
Total Orders	1,80,519
Total Columns	47 Features
Data Type	Transactional Supply Chain Data
Scope	Global (Multi-Market, Multi-Region)

Column Categories Breakdown:-

1. Customer Information (8 columns)
Customer Id, Customer Fname, Customer Lname, Customer City, Customer Country, Customer State, Customer Street, Customer Zipcode
2. Order Details (10 columns)
Order Id, Order Date, Order Status, Order City, Order Country, Order Region, Order State, Order Customer Id, Type, Order Item Id
3. Shipping & Delivery (6 columns)
Shipping Date, Shipping Mode, Shipping Delay (days), Days for shipping (real), Days for shipment (scheduled), Delivery Status, Late_delivery_risk
4. Financial Metrics (8 columns)
Sales, Sales per customer, Benefit per order, Order Item Discount, Order Item Discount Rate, Order Item Product Price, Order Item Profit Ratio, Order Item Total, Order Profit Per Order
5. Product Information (5 columns)
Product Card Id, Product Name, Product Price, Product Category Id, Order Item Cardprod Id, Order Item Quantity
6. Category & Department (4 columns)
Category Id, Category Name, Department Id, Department Name
7. Geographic & Market (3 columns)
Market, Latitude, Longitude

2.2 Business Context

Factor	Description
 Global Operations	Orders span multiple countries, markets & regions worldwide
 Diverse Customers	Multiple customer segments with varying buying behaviors
 Multi-Category Products	Various departments & product categories involved
 Complex Logistics	Multiple shipping modes with real vs scheduled delivery gaps
 Financial Pressure	Discounts, profit ratios & order benefits need close monitoring

2.3 Problem Statement

Despite having large volumes of order data, organizations often struggle to translate raw supply chain data into actionable business intelligence — leading to undetected delivery failures, profit leakages, and missed growth opportunities."

The key business problems identified in this project are:

- **High Late Delivery Risk** — A significant portion of orders are flagged with late delivery risk, impacting customer satisfaction
- **Profitability Gaps** — Discounts and order item pricing are not always aligned with actual profit generation
- **Shipping Inefficiencies** — Real shipping days frequently deviate from scheduled days
- **Market Imbalance** — Revenue and profit are unevenly distributed across global markets
- **Customer Insight Gap** — Limited visibility into which customer segments drive the most value

2.4 Why This Analysis Matters

Business Impact of Supply Chain Inefficiencies:

Late Deliveries → **Poor Customer Experience & Churn**

High Discounts → **Reduced Profit Margins**

Wrong Shipping Mode → **Increased Logistics Costs**

Low-Profit Products → **Revenue Leakage**

Market Blind Spots → **Missed Growth Opportunities**

☒ **Data-Driven Analysis** → Smarter Decisions & Better Outcomes

Exploratory Data Analysis (EDA) using Python:

	count	mean	std	min	25%	50%	75%	max
Days for shipping (real)	180519.0	3.497654	1.623722	0.000000	2.000000	3.000000	5.000000	6.000000
Days for shipment (scheduled)	180519.0	2.931847	1.374449	0.000000	2.000000	4.000000	4.000000	4.000000
Benefit per order	180519.0	21.974989	104.433526	-4274.979980	7.000000	31.520000	64.800003	911.799988
Sales per customer	180519.0	183.107609	120.043670	7.490000	104.379997	163.990005	247.399994	1939.989990
Late_delivery_risk	180519.0	0.548291	0.497664	0.000000	0.000000	1.000000	1.000000	1.000000
Category Id	180519.0	31.851451	15.640064	2.000000	18.000000	29.000000	45.000000	76.000000
Customer Id	180519.0	6691.379495	4162.918106	1.000000	3258.500000	6457.000000	9779.000000	20757.000000
Customer Zipcode	180516.0	35921.126914	37542.461122	603.000000	725.000000	19380.000000	78207.000000	99205.000000
Department Id	180519.0	5.443460	1.629246	2.000000	4.000000	5.000000	7.000000	12.000000
Latitude	180519.0	29.719955	9.813646	-33.937553	18.265432	33.144863	39.279617	48.781933
Longitude	180519.0	-84.915675	21.433241	-158.025986	-98.446312	-76.847908	-66.370583	115.263077
Order Customer Id	180519.0	6691.379495	4162.918106	1.000000	3258.500000	6457.000000	9779.000000	20757.000000
Order Id	180519.0	36221.894903	21045.379569	1.000000	18057.000000	36140.000000	54144.000000	77204.000000
Order Item Cardprod Id	180519.0	692.509764	336.446807	19.000000	403.000000	627.000000	1004.000000	1363.000000
Order Item Discount	180519.0	20.664741	21.800901	0.000000	5.400000	14.000000	29.990000	500.000000
Order Item Discount Rate	180519.0	0.101668	0.070415	0.000000	0.040000	0.100000	0.160000	0.250000

Order Item Id	180519.0	90260.000000	52111.490959	1.000000	45130.500000	90260.000000	135389.500000	180519.000000
Order Item Product Price	180519.0	141.232550	139.732492	9.990000	50.000000	59.990002	199.990005	1999.989990
Order Item Profit Ratio	180519.0	0.120647	0.466796	-2.750000	0.080000	0.270000	0.360000	0.500000
Order Item Quantity	180519.0	2.127638	1.453451	1.000000	1.000000	1.000000	3.000000	5.000000
Sales	180519.0	203.772096	132.273077	9.990000	119.980003	199.919998	299.950012	1999.989990
Order Item Total	180519.0	183.107609	120.043670	7.490000	104.379997	163.990005	247.399994	1939.989990
Order Profit Per Order	180519.0	21.974989	104.433526	-4274.979980	7.000000	31.520000	64.800003	911.799988
Product Card Id	180519.0	692.509764	336.446807	19.000000	403.000000	627.000000	1004.000000	1363.000000
Product Category Id	180519.0	31.851451	15.640064	2.000000	18.000000	29.000000	45.000000	76.000000
Product Price	180519.0	141.232550	139.732492	9.990000	50.000000	59.990002	199.990005	1999.989990
Shipping Delay (days)	180519.0	0.565807	1.490966	-2.000000	0.000000	1.000000	1.000000	4.000000

Descriptive Statistics — Key Insights

Shipping & Delivery

- **Days for Shipping** (real) averages 3.5 days vs Days for Shipment (scheduled) at 2.9 days — meaning orders consistently arrive ~0.57 days later than scheduled, confirmed by the Shipping Delay (days) mean of 0.57.
- **Shipping delay ranges** from -2 days (early delivery) to +4 days (severely late), showing high variability in fulfillment performance.
- **Late Delivery Risk mean is 0.548** — meaning roughly 54.8% of all orders carry a late delivery risk, which is a significant operational concern.

Sales & Revenue

- Sales range from **\$9.99 to \$1,999.99** with a mean of **\$203.77** and **std of \$132.27** — indicating a wide spread of order values across the dataset.
- Order Item Total closely mirrors Sales (**mean \$183.10**), suggesting most orders are single-item purchases.
- Sales per Customer averages **\$183.10** with a max of **\$1,939.99**, showing a small segment of high-value customers.

Profitability

- Benefit per Order has a mean of **\$21.97** but a minimum of **-\$4,274.98** — indicating severely loss-making orders exist in the dataset.
- Order Profit Per Order also shows the same range (**-\$4,274.98 to \$911.80**), confirming significant profit leakage in a portion of orders.
- Order Item Profit Ratio averages only **0.12 (12%)** with values as low as **-2.75** — meaning some products are sold at nearly 3x their cost as a loss.
- **75%** of orders have a profit ratio below **0.36**, suggesting most transactions operate on thin margins.

Discount Analysis

- Order Item Discount has a mean of **\$20.66** but a max of **\$500** — extreme discounting is happening on select orders.
- Order Item Discount Rate averages **10.2%** with a max of **25%**, showing discounts are capped at quarter price but not always — the **\$500** max discount suggests data anomalies worth investigating.

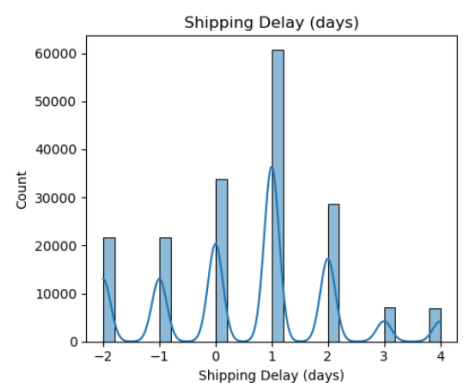
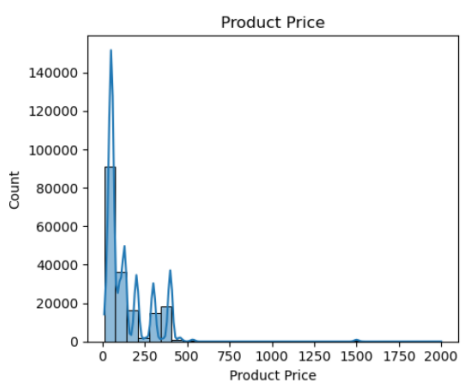
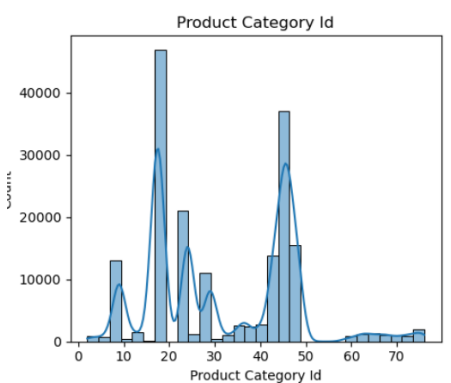
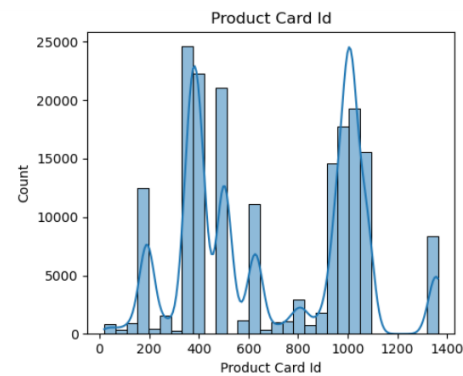
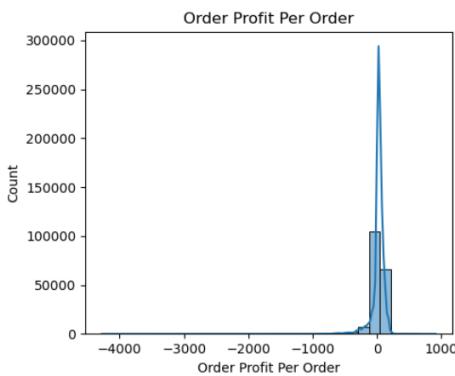
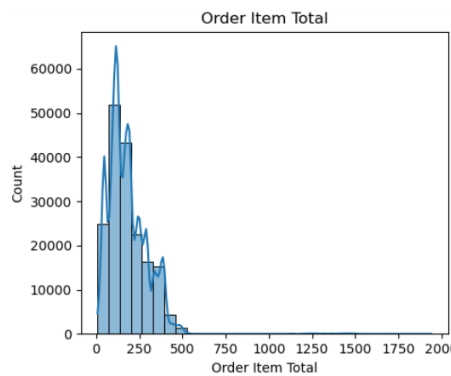
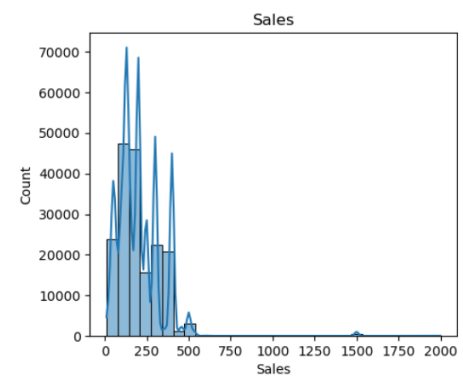
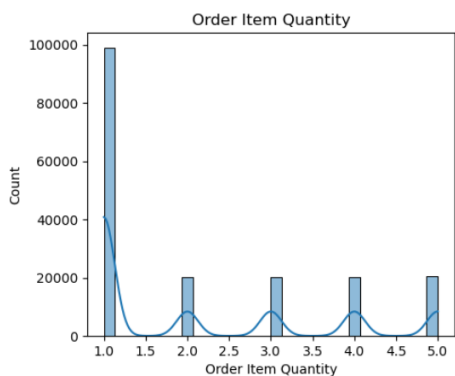
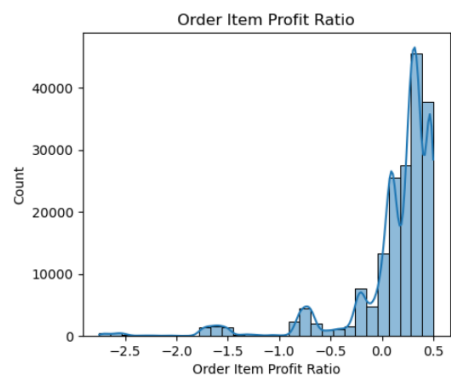
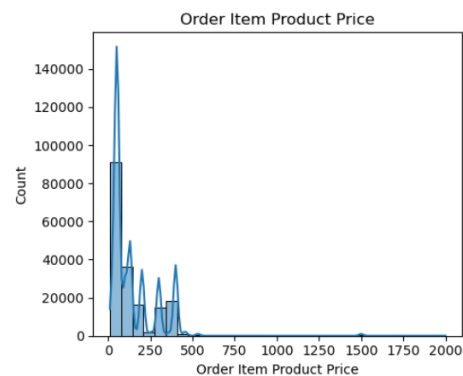
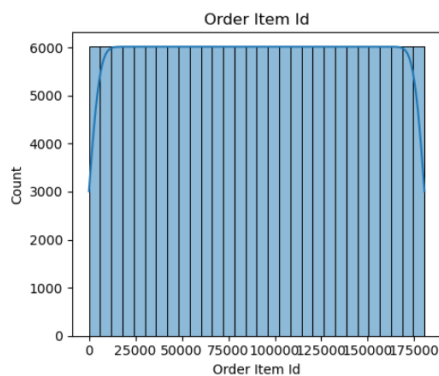
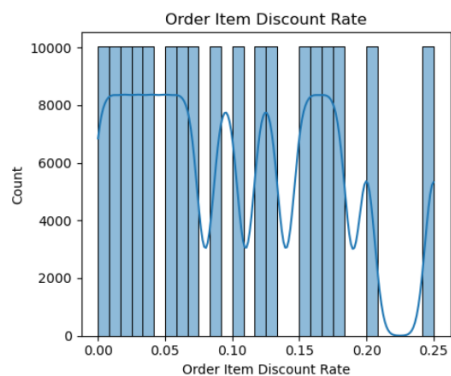
Product & Pricing

- Order Item Product Price and Product Price are identical (**mean \$141.23, max \$1,999.99**) — consistent product catalog pricing.
- Order Item Quantity averages just **2.13** units per order (max 5), indicating most customers buy in small quantities.
- Product Category Id ranges from 2 to 76 with mean **31.85** — a wide variety of product categories are present.

Geographic

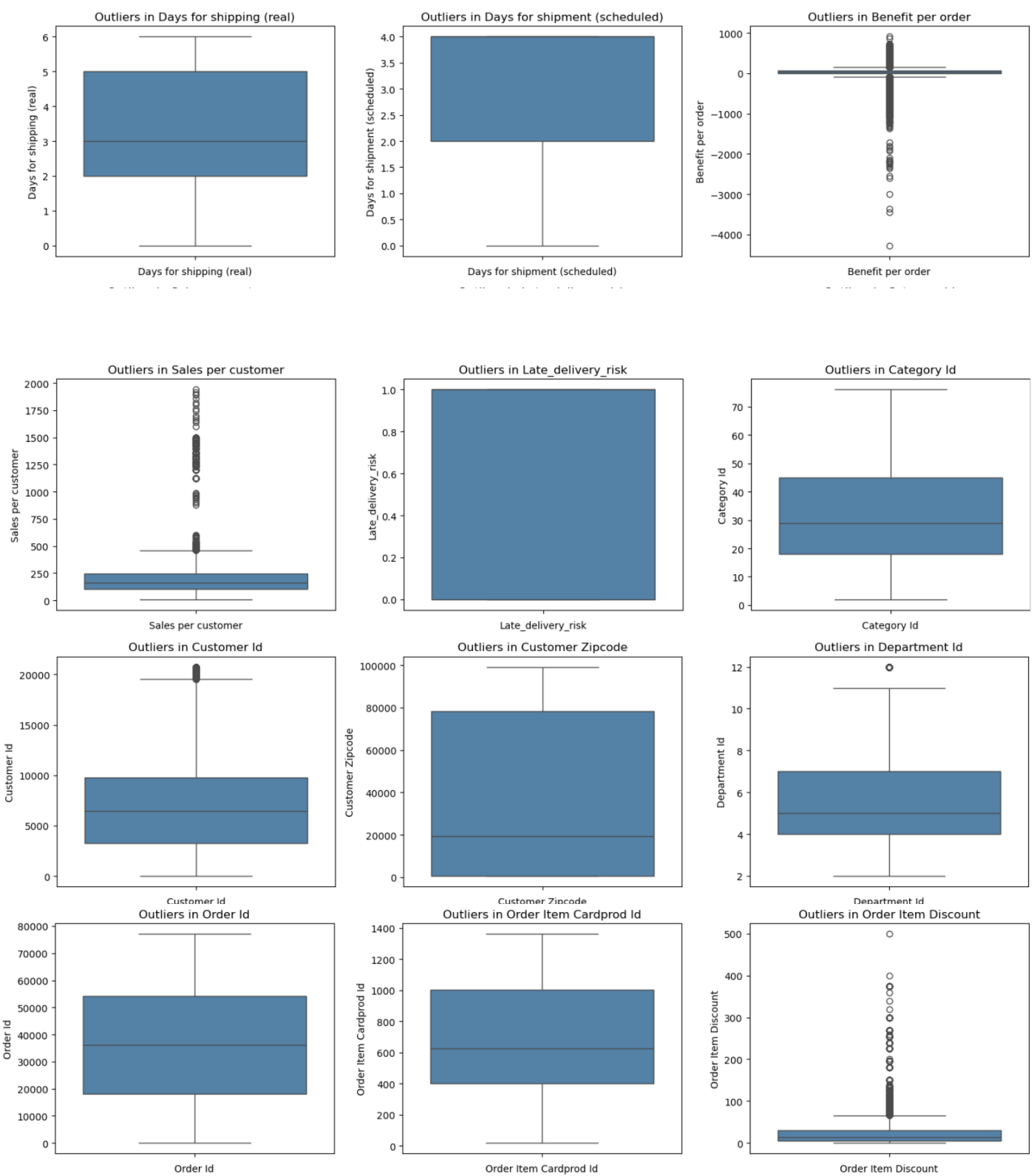
- Latitude ranges from **-33.94 to +48.78** covering South America to Europe/North America.
- Longitude spans **-158.03 to +115.26** — confirming truly global operations across multiple continents.

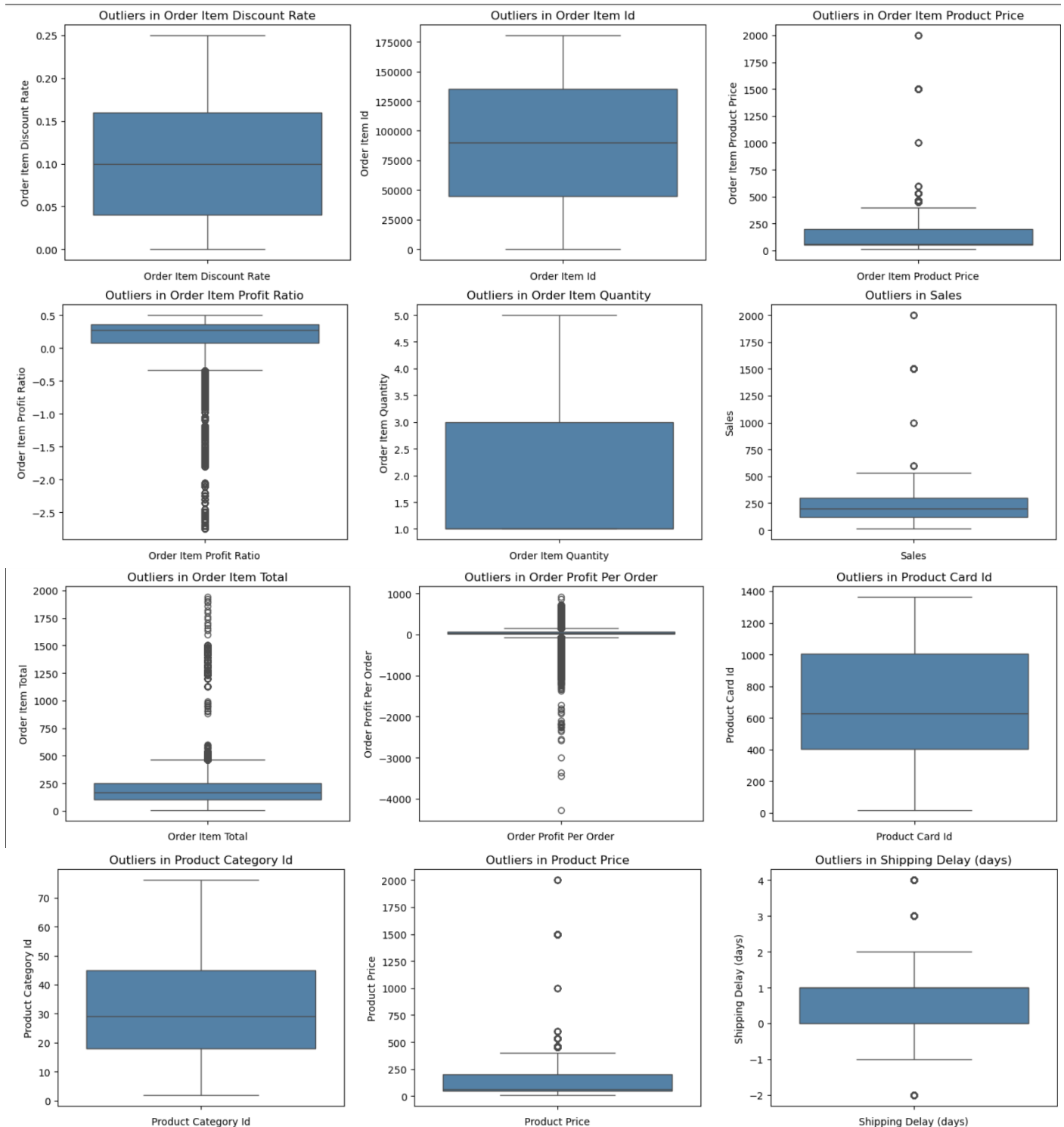




Outlier Detection:

Outlier Detection - Boxplots





Outlier Analysis:

◆ Overview

The outlier analysis was conducted across all numerical columns in the supply chain dataset of 1,80,519 orders using the IQR method to identify anomalies affecting data quality and business performance.

◆ **Clean Columns :** A total of 12 columns returned zero outliers, including Days for Shipping, Late Delivery Risk, Order Item Quantity, and Order Item Discount Rate, confirming these fields are consistently bounded and structurally clean across the entire dataset.

◆ Critical Outliers:

Shipping Delay (days) recorded the highest outlier count at 35,701 — nearly 19.8% of all orders, indicating a systemic delivery execution failure directly linked to the 54.8% late delivery risk rate identified in descriptive statistics.

◆ Profitability Outliers:

Benefit per Order and Order Profit Per Order each contain **18,942** outliers, with profits dropping as low as **-\$4,274.98**, revealing deep loss-making transactions and significant revenue leakage. Order Item Profit Ratio follows with **17,300** outliers, pointing to widespread negative margins at the product level.

◆ Discount Outliers

7,537 orders carry abnormal discount values reaching up to **\$500**, suggesting uncontrolled discounting practices that are actively eroding profit margins without clear strategic justification.

◆ Medium Priority Outliers

Moderate outliers were detected in Sales, Product Price, Customer Id, and Geographic coordinates. These largely reflect premium customers, high-value products, and remote delivery locations rather than data entry errors, and should be retained for segmentation analysis.

Key Performance Indicators (KPIs)

- **On-Time Delivery Rate:** 40.88% of total orders were delivered within the expected delivery schedule, indicating the efficiency of the logistics and transportation process.
- **Late Delivery Rate:** 54.83% of orders experienced delivery delays, highlighting potential issues in shipping operations, supplier coordination, or route management.
- **Average Shipping Delay:** Orders were delayed by an average of **0.57 days**, showing moderate delivery disruptions across the supply chain network.
- **Total Revenue:** The business generated a total revenue of **\$36.78 Million**, reflecting strong overall sales performance during the analysis period.
- **Total Profit:** The company achieved a total profit of **\$3.97 Million**, demonstrating positive business profitability after operational costs.
- **Overall Profit Margin:** The overall profit margin stood at **10.78%**, indicating the percentage of revenue converted into actual profit.

- **Orders at a Loss:** Approximately **18.71%** of total orders resulted in financial losses, suggesting the impact of high discounts, shipping costs, or low-margin products.
- **Average Order Value (AOV):** The average value per customer order was **\$559.45**, representing the average customer spending per transaction.
- **Order Fulfillment Rate:** The organization successfully fulfilled **43.82%** of orders without operational issues, reflecting supply chain execution efficiency.
- **Average Items per Order:** Customers purchased an average of **5.84 items** per order, indicating moderate basket size and purchasing behavior.
- **Late Delivery Risk Rate:** Around **54.83%** of orders were identified as high-risk for delayed delivery, emphasizing the need for logistics optimization and better demand planning.
- **Average Discount Rate:** The company provided an average discount of **10.17%** across all orders, which may have contributed to increased sales volume but affected profitability.
- **Average Profit Ratio per Item:** Each product generated an average profit ratio of **12.06%**, reflecting the profitability performance at the item level.

DELIVERY & SHIPPING ANALYSIS:

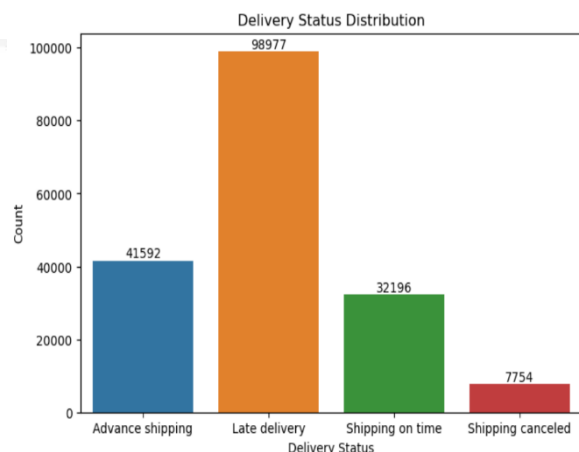
Delivery Status Analysis:

- **Late Delivery:** A total of **98,977 orders** were delivered late, representing the largest share of all delivery statuses.

This indicates significant challenges in transportation efficiency, inventory planning, or supply chain coordination.

- **Advance Shipping:** Around **41,592 orders** were shipped earlier than the scheduled delivery date, reflecting efficient logistics management and proactive order processing for a portion of customers.
- **Shipping On Time:** Approximately **32,196 orders** were delivered on schedule, demonstrating the company's ability to maintain timely deliveries for a moderate number of orders.
- **Shipping Canceled:** A total of **7,754 orders** were canceled before delivery, which may be due to stock shortages, customer cancellations, operational inefficiencies, or supplier-related issues.

Delivery_status	orders
Late delivery	98977
Advance shipping	41592
Shipping on time	32196
Shipping canceled	7754



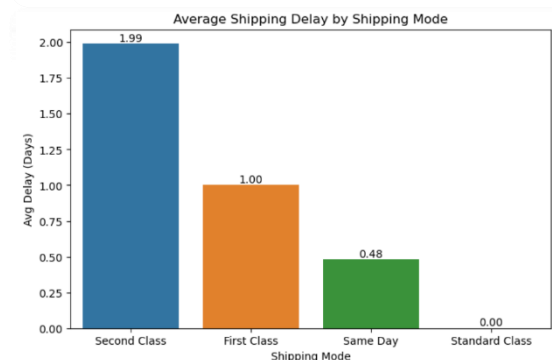
Shipping Delay Statistics Analysis

- **Average Shipping Delay (Mean):** The average shipping delay was **0.57 days**, indicating that most orders experienced only minor delays during the delivery process.
- **Median Shipping Delay:** The median delay was **1 day**, which means that at least half of the delayed shipments were delivered within one day beyond the expected delivery date.
- **Maximum Shipping Delay:** The highest recorded shipping delay was **4 days**, showing that some orders faced significant delivery disruptions.
- **Orders with Delay Greater Than 0 Days:** Approximately **57.3%** of total orders experienced delivery delays, indicating that more than half of the shipments were not delivered on time.

Shipping mode performance:

- **Second Class Shipping:** Second Class deliveries recorded the highest average delay of **1.99 days**, indicating slower transportation performance and a greater risk of delayed deliveries compared to other shipping methods.
- **First Class Shipping:** First Class shipments experienced an average delay of **1 day**, showing relatively better delivery efficiency but still facing moderate shipping delays.
- **Same Day Shipping:** Same Day delivery had a very low average delay of **0.48 days**, reflecting strong operational efficiency and faster order fulfilment performance.
- **Standard Class Shipping:** Standard Class shipping showed an average delay close to **0 days**, indicating stable and consistent delivery performance with minimal delays.

shipping_mode	avg_shipping_delay_days
Second Class	1.99
First Class	1.00
Same Day	0.48
Standard Class	0.00



SALES & PROFIT ANALYSIS:

Sales Performance Analysis

- **Total Sales:** The business generated total sales revenue of **\$36.78 Million**, reflecting strong market demand and overall business growth during the analysis period.
- **Average Sales per Order:** The average sales value per order was **\$203.77**, indicating the average amount spent by customers in each transaction.

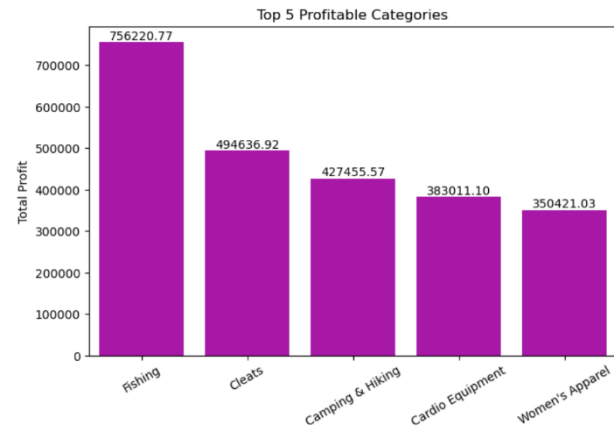
Profitability Analysis

- **Total Profit:** The company generated a total profit of **\$3.97 Million**, indicating strong overall financial performance and successful revenue generation after covering operational and supply chain costs.
- **Average Profit per Order:** The average profit earned per order was **\$21.97**, reflecting the profitability contribution of each customer transaction.
- **Orders at Loss:** Approximately **18.7%** of total orders resulted in a financial loss, suggesting that a significant portion of transactions were negatively impacted by factors such as high discounts, shipping expenses, operational costs, or low-margin products.

Top 5 Most Profitable Product Categories

- Fishing:** The Fishing category generated the highest profit of **\$756,220.77**, making it the most profitable product segment in the business portfolio.
- Cleats:** Cleats contributed a total profit of **\$494,636.92**, reflecting strong customer demand and healthy product margins within the sports footwear segment.
- Camping & Hiking:** The Camping & Hiking category earned **\$427,455.57** in profit, indicating strong performance in outdoor and recreational products.
- Cardio Equipment:** Cardio Equipment generated a total profit of **\$383,011.10**, showing consistent profitability in the fitness and health equipment market.
- Women's Apparel:** Women's Apparel produced **\$350,421.03** in profit, highlighting stable customer demand and solid revenue contribution from the fashion segment.

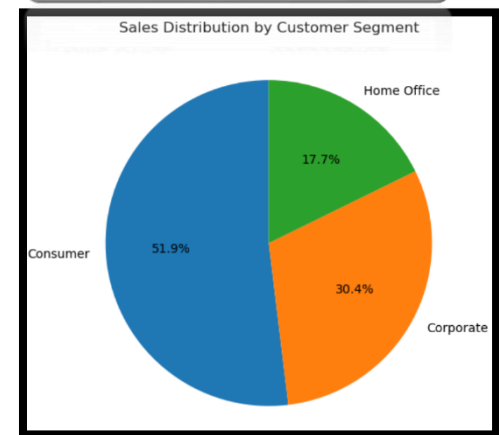
category_name	Profit
Fishing	756220.77
Cleats	494636.92
Camping & Hiking	427455.57
Cardio Equipment	383011.1
Women's Apparel	350421.03



Sales Analysis by Customer Segment

- Consumer Segment:** The Consumer segment generated the highest sales revenue of **\$19.10 Million**, making it the largest contributor to overall business sales. This indicates strong purchasing activity from individual retail customers.
- Corporate Segment:** The Corporate segment contributed **\$11.17 Million** in sales, reflecting significant demand from business and organizational customers.
- Home Office Segment:** The Home Office segment generated **\$6.52 Million** in sales, showing moderate purchasing activity from small businesses and remote office customers.

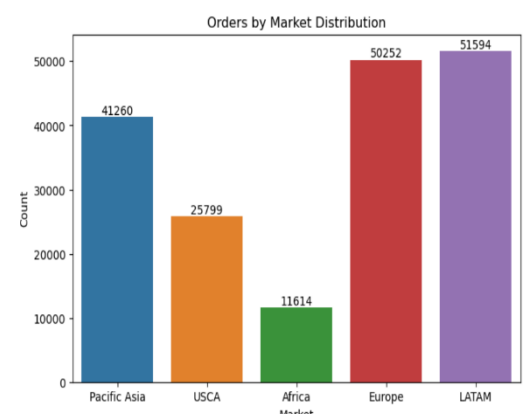
customer_segment	Revenue
Consumer	19095790.16
Corporate	11168406.84
Home Office	6520538.02



GEOGRAPHIC ANALYSIS:

Market-Wise Order Distribution Analysis

- LATAM Market:** The LATAM region recorded the highest number of orders with **51,594 orders**, indicating strong customer demand and a significant market presence in Latin America.
- Europe Market:** Europe contributed **50,252 orders**, making it the second-largest market with consistently high sales and customer activity.



- **Pacific Asia Market:** The Pacific Asia region generated **41,260 orders**, reflecting strong market performance and growing demand across Asian-Pacific countries.
- **USCA Market:** The USCA market accounted for **25,799 orders**, showing moderate sales contribution compared to larger international markets.
- **Africa Market:** Africa recorded **11,614 orders**, representing the smallest order volume among the listed markets but still contributing to the company's global supply chain operations.

market	orders
LATAM	51594
Europe	50252
Pacific Asia	41260
USCA	25799
Africa	11614

Market-Wise Sales & Profit:

- **Africa:** The Africa market processed **11,614 orders**, generating total revenue of **\$2.29 Million** and profit of **\$0.25 Million**. It recorded the lowest market contribution among all regions.
- **Europe:** Europe generated the highest revenue of **\$10.87 Million** with a total profit of **\$1.17 Million** from **50,252 orders**. This makes Europe the strongest-performing market in the business.
- **LATAM:** The LATAM market handled **51,594 orders**, contributing **\$10.28 Million** in revenue and **\$1.12 Million** in profit. It recorded the highest number of total orders among all markets.
- **Pacific Asia:** Pacific Asia processed **41,260 orders**, generating **\$8.27 Million** in revenue and **\$0.86 Million** in profit, reflecting strong business performance in the Asia-Pacific region.
- **USCA:** The USCA market recorded **25,799 orders**, with total revenue of **\$5.07 Million** and profit of **\$0.56 Million**. It maintained stable sales and profitability performance.
- Overall, the company processed **180,519 orders**, generating total revenue of **\$36.78 Million** and profit of **\$3.97 Million**. Europe and LATAM were the top-performing markets, contributing the highest share of revenue and profit across global operations.

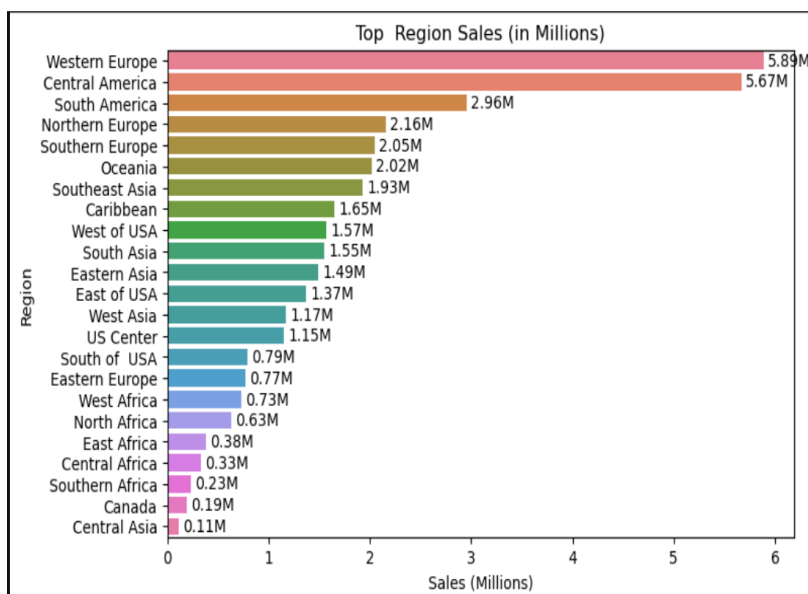
Market Performance				
Market	Orders	Revenue	Profit	
Africa	11614	\$2.29 M	\$0.25 M	
Europe	50252	\$10.87 M	\$1.17 M	
LATAM	51594	\$10.28 M	\$1.12 M	
Pacific Asia	41260	\$8.27 M	\$0.86 M	
USCA	25799	\$5.07 M	\$0.56 M	
Grand Total	180519	\$36.78 M	\$3.97 M	

Region-Wise Sales Performance Analysis

- **Western Europe:** Western Europe generated the highest sales revenue of **\$5.89 Million**, making it the top-performing region in the global market.
- **Central America:** Central America recorded **\$5.67 Million** in sales, showing strong customer demand and a significant contribution to overall revenue.
- **South America:** South America contributed **\$2.96 Million** in sales, reflecting solid market performance and growing business opportunities within the region.
- **Northern Europe:** Northern Europe achieved **\$2.16 Million** in sales, demonstrating stable demand and consistent regional performance.
- **Southern Europe:** Southern Europe generated **\$2.05 Million** in revenue, indicating healthy sales activity across European markets.

Order Region	
Western Europe	5.89M
Central America	5.67M
South America	2.96M
Northern Europe	2.16M
Southern Europe	2.05M
Oceania	2.02M
Southeast Asia	1.93M
Caribbean	1.65M
West of USA	1.57M
South Asia	1.55M
Eastern Asia	1.49M
East of USA	1.37M
West Asia	1.17M
US Center	1.15M
South of USA	0.79M
Eastern Europe	0.77M
West Africa	0.73M
North Africa	0.63M
East Africa	0.38M
Central Africa	0.33M
Southern Africa	0.23M
Canada	0.19M
Central Asia	0.11M

- **Oceania:** Oceania recorded **\$2.02 Million** in sales, highlighting steady business growth in the region.
- **Southeast Asia:** Southeast Asia contributed **\$1.93 Million** in revenue, reflecting increasing customer demand and expanding market potential.
- **Caribbean:** The Caribbean region generated **\$1.65 Million** in sales, showing moderate market performance.
- **West of USA:** The western region of the USA recorded **\$1.57 Million** in sales, contributing significantly to domestic market revenue.
- **South Asia:** South Asia generated **\$1.55 Million** in sales, indicating growing regional demand and market expansion opportunities.

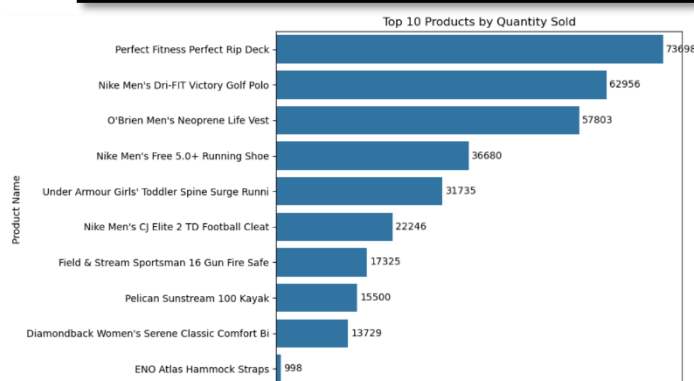


PRODUCT ANALYSIS:

Top-Selling Products Analysis:

The product analysis shows that **Perfect Fitness Perfect Rip Deck** recorded the highest sales with **73,698 units**, followed by **Nike Men's Dri-FIT Victory Golf Polo** and **O'Brien Men's Neoprene Life Vest**. Sportswear, fitness, and outdoor recreation products dominate overall sales performance, indicating strong customer demand in these categories. Products from brands like Nike and Under Armour contribute significantly to revenue generation. Athletic footwear, fitness equipment, and outdoor products remain key drivers of business growth. Maintaining inventory availability and targeted marketing for these high-performing products can further improve sales and profitability.

product_name	Total_order_quantity
Perfect Fitness Perfect Rip Deck	73698
Nike Men's Dri-FIT Victory Golf Polo	62956
O'Brien Men's Neoprene Life Vest	57803
Nike Men's Free 5.0+ Running Shoe	36680
Under Armour Girls' Toddler Spine Surge Runni	31735
Nike Men's CJ Elite 2 TD Football Cleat	22246
Field & Stream Sportsman 16 Gun Fire Safe	17325
Pelican Sunstream 100 Kayak	15500
Diamondback Women's Serene Classic Comfort Bi	13729
ENO Atlas Hammock Straps	998



Top 10 Products by Sales Revenue:

The sales analysis shows that **Field & Stream Sportsman 16 Gun Fire Safe** generated the highest revenue of **\$6.93 Million**, making it the top-performing product. **Perfect Fitness Perfect Rip Deck** and **Diamondback Women's Serene Classic Comfort Bike** also contributed significantly with revenues above **\$4 Million**. Sportswear, fitness equipment, and outdoor recreation products dominate the top-selling product list, highlighting strong customer demand in these categories. Popular brands such as Nike, Under

Top 10 Product By Sales	
Product Name	Revenue
Field & Stream Sportsman 16 Gun Fire Safe	6929653.69
Perfect Fitness Perfect Rip Deck	4421143.14
Diamondback Women's Serene Classic Comfort Bi	4118425.57
Nike Men's Free 5.0+ Running Shoe	3667633.20
Nike Men's Dri-FIT Victory Golf Polo	3147800.00
Pelican Sunstream 100 Kayak	3099845.09
Nike Men's CJ Elite 2 TD Football Cleat	2891757.66
O'Brien Men's Neoprene Life Vest	2888993.91
Under Armour Girls' Toddler Spine Surge Runni	1269082.67
Dell Laptop	663000.00

Armour, and Dell play a major role in overall revenue generation. Focusing on inventory management, product availability, and targeted promotions for these high-performing products can further increase sales and profitability.

Minimum Selling Products:

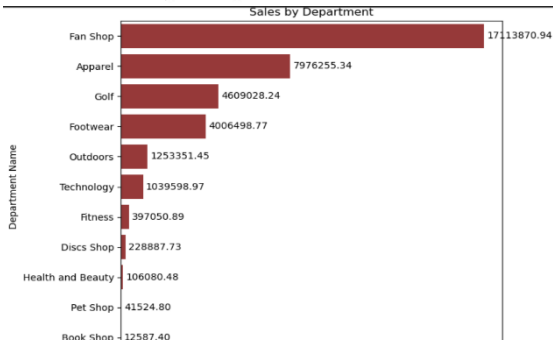
Bowflex SelectTech 1090 Dumbbells generated the lowest revenue at **\$5,999.90**, while **Bushnell Pro X7 Jolt Slope Rangefinder** recorded a negative profit of **-\$255.95**, indicating a loss-making product. Other low-performing products include golf accessories, sportswear, fitness equipment, and toys with limited revenue generation. Despite low sales, some products such as **LIJA Women’s Button Golf Dress** and **Merrell Women’s Grassbow Sport Waterproof Hiking Shoes** maintained positive profit margins. The company should review pricing strategies, customer demand, inventory levels, and promotional activities for these underperforming products to improve profitability and reduce financial losses.

Min Saling Product		
Product Name	Revenue	Profit
Bowflex SelectTech 1090 Dumbbells	5999.90	1190.78
Bushnell Pro X7 Jolt Slope Rangefinder	6599.89	-255.95
CDs of rock	3059.59	383.85
Cleveland Golf Women's 588 RTX CB Satin Chrom	8399.30	370.61
Diamondback Girls' Clarity 24 Hybrid Bike 201	8399.72	284.42
LIJA Women's Button Golf Dress	8208.00	1156.60
Merrell Women's Grassbow Sport Waterproof Hik	7539.42	1029.71
Nike Men's Fingertrap Max Training Shoe	7999.36	602.38
Polar Loop Activity Tracker	5937.30	326.20
Toys	6104.66	900.71

Department-Wise Sales

the **Fan Shop** department generated the highest sales revenue of **\$17.11 Million**, making it the largest contributor to overall business performance. The **Apparel** department followed with **\$7.98 Million** in sales, while the **Golf** and **Footwear** departments also performed strongly with revenues above **\$4 Million**. Departments related to sports, clothing, and outdoor activities dominate overall sales, indicating high customer demand in these categories. Smaller departments such as **Pet Shop** and **Book Shop** contributed comparatively lower revenue, showing limited market performance. Focusing on high-performing departments while improving marketing and product strategies for low-performing segments could help increase overall business growth and profitability.

department_name	Revenue
Fan Shop	17113871
Apparel	7976255
Golf	4609028
Footwear	4006499
Outdoors	1253351
Technology	1039599
Fitness	397051
Discs Shop	228888
Health and Beauty	106080
Pet Shop	41525
Book Shop	12587



Top 10 Category Sales & Profit:

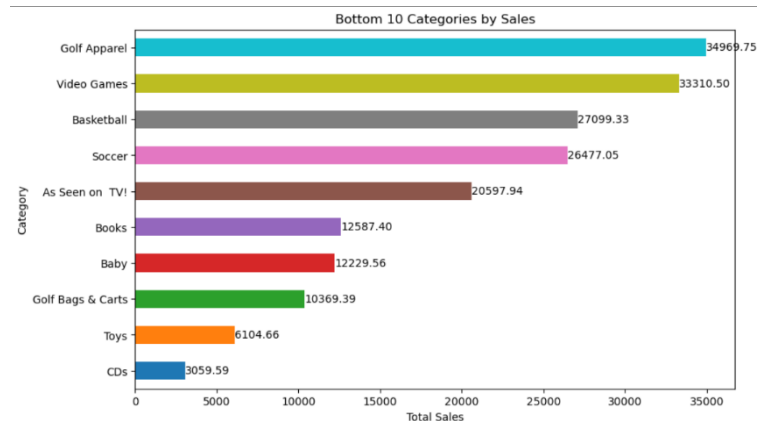
The category analysis shows that **Fishing** generated the highest revenue of **\$6.93 Million** and the highest profit of **\$756,220.77**, making it the best-performing category overall. **Cleats**, **Camping & Hiking**, and **Cardio Equipment** also contributed significantly to both sales and profitability. Sports, fitness, outdoor recreation, and apparel categories dominate the top-performing segments, indicating strong customer demand in these markets. Categories such as **Water Sports**, **Men’s Footwear**, and **Women’s Apparel** maintained stable revenue and profit performance throughout the analysis period. In comparison,

Top 10 Category Sales & Profit		
Category Name	Revenue	Total Profit
Camping & Hiking	4118425.57	427455.57
Cardio Equipment	3694843.20	383011.10
Cleats	4431942.78	494636.92
Computers	663000.00	69656.81
Fishing	6929653.69	756220.77
Indoor/Outdoor Games	2888993.91	318451.43
Men's Footwear	2891757.66	311902.82
Shop By Sport	1309522.04	129813.96
Water Sports	3113844.68	325146.96
Women's Apparel	3147800.00	350421.03
Grand Total	33189783.54	3566717.36

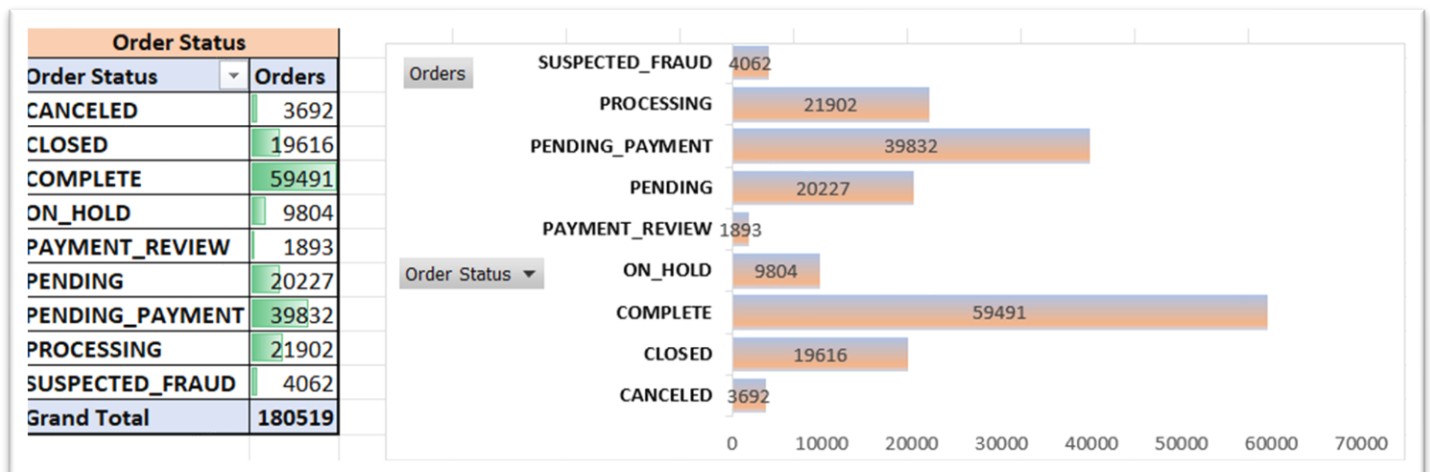
the **Computers** category generated the lowest revenue and profit among the top 10 categories, suggesting comparatively lower market demand.

Minimum Revenue Categories:

minimum revenue categories shows that **CDs** generated the lowest sales revenue of **\$3,059.59**, followed by **Toys** with **\$6,104.66** and **Golf Bags & Carts** with **\$10,369.39**. Other low-performing categories include **Baby** (**\$12,229.56**), **Books** (**\$12,587.40**), and **As Seen on TV!** products (**\$20,597.94**). Categories such as **Soccer**, **Basketball**, **Video Games**, and **Golf Apparel** also recorded comparatively lower sales performance. These results indicate weaker customer demand and lower market contribution from these categories compared to high-performing product segments.



Order Status:



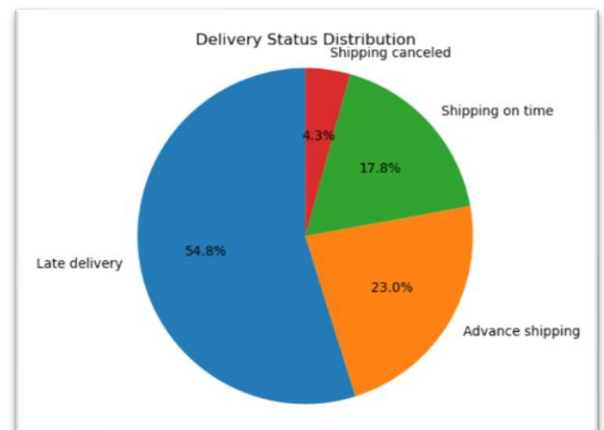
- **CANCELED:** A total of **3,692 orders** were canceled, indicating a relatively small portion of total transactions affected by customer cancellations, inventory shortages, or operational issues.
- **CLOSED:** The company recorded **19,616 closed orders**, representing successfully completed order cycles after fulfillment and final processing.
- **COMPLETE:** The **COMPLETE** status accounted for the highest number of orders with **59,491 orders**, showing strong order fulfillment performance and successful transaction completion.
- **ON_HOLD:** Around **9,804 orders** were placed on hold, possibly due to payment verification, stock availability, or delivery-related issues.
- **PAYMENT_REVIEW:** A total of **1,893 orders** remained under payment review, indicating pending payment verification or transaction approval processes.
- **PENDING:** The business recorded **20,227 pending orders**, reflecting orders awaiting processing, confirmation, or shipment.
- **PENDING_PAYMENT:** Approximately **39,832 orders** were marked as pending payment, highlighting a significant number of incomplete payment transactions.
- **PROCESSING:** The **PROCESSING** status included **21,902 orders**, representing transactions currently under fulfillment or shipment preparation.

- **SUSPECTED_FRAUD:** A total of **4,062 orders** were flagged as suspected fraud, indicating potential risks related to payment security or suspicious customer activity.
- **Overall Insight:** Out of **180,519 total orders**, the majority were successfully completed, while a large number of orders remained in pending or processing stages. The high volume of pending payment orders suggests opportunities to improve payment systems, customer communication, and transaction verification processes.

Delivery Status:

- **Advance Shipping:** Advance shipping accounted for **41,592 orders**, generating total revenue of **\$8.52 Million** and profit of **\$935.23 Thousand**. This indicates strong operational efficiency and timely shipment processing for early deliveries.
- **Late Delivery:** Late deliveries represented the highest number of orders at **98,977**, contributing **\$20.13 Million** in revenue and **\$2.14 Million** in profit. Although delayed shipments generated the largest share of revenue, the high delay rate highlights major logistics and supply chain performance issues.
- **Shipping Canceled:** A total of **7,754 orders** were canceled, resulting in revenue of **\$1.57 Million** and profit of **\$160.48 Thousand**. Order cancellations may be linked to stock shortages, operational inefficiencies, or customer-related issues.
- **Shipping On Time:** On-time deliveries included **32,196 orders**, generating revenue of **\$6.57 Million** and profit of **\$731.14 Thousand**. This reflects moderate delivery efficiency and stable customer service performance.

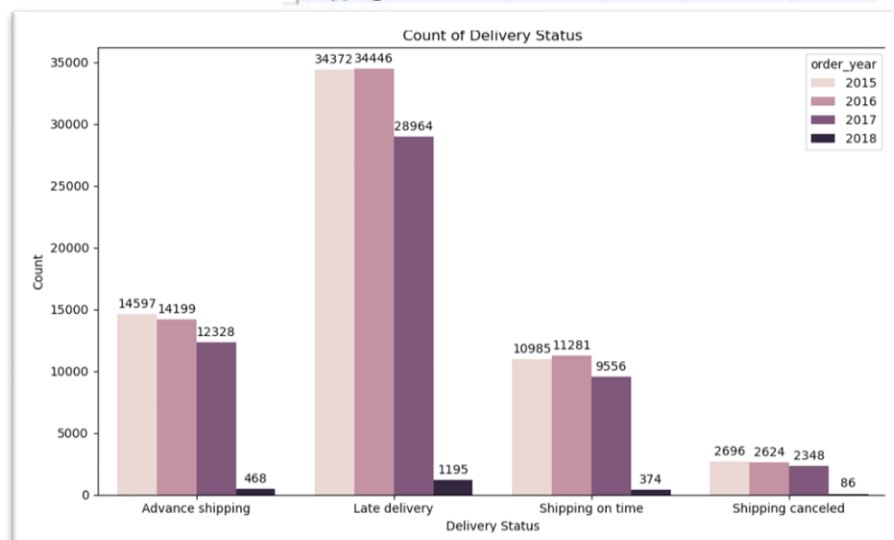
Delivery Status			
Delivery Status	Orders	Revenue	Profit
Advance shipping	41592	8518007.891	935225.2811
Late delivery	98977	20126395.27	2140051.682
Shipping canceled	7754	1570305.36	160482.3402
Shipping on time	32196	6570026.497	731143.671
Grand Total	180519	36784735.01	3966902.974



Late Delivery Status Year Wise:

- **Advance Shipping:** Advance shipping orders remained relatively stable from 2015 to 2017, with **14,597 orders in 2015**, **14,199 in 2016**, and **12,328 in 2017**. In 2018, the count sharply declined to **468 orders**, likely due to incomplete yearly data.
- **Late Delivery:** Late deliveries consistently recorded the highest number of orders each year, with **34,372 orders in 2015**, **34,446 in 2016**, and **28,964 in 2017**. This highlights ongoing logistics and

Delivery_Status	y2015	y2016	y2017	y2018
Advance shipping	14597	14199	12328	41592
Late delivery	34372	34446	28964	98977
Shipping canceled	2696	2624	2348	7754
Shipping on time	10985	11281	9556	32196

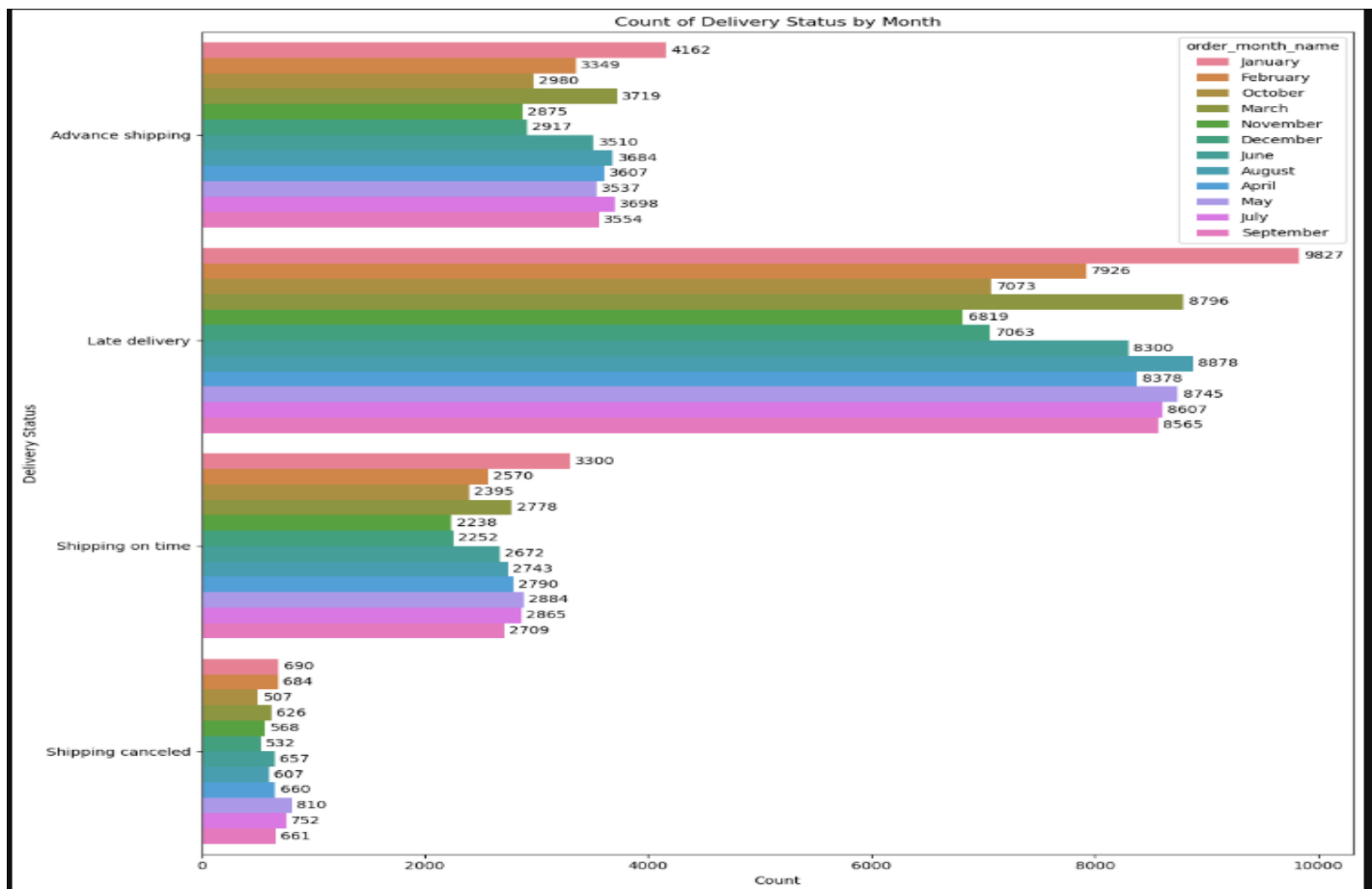


transportation challenges within the supply chain.

- **Shipping Canceled:** Canceled shipments remained comparatively low across all years, with **2,696 orders in 2015, 2,624 in 2016, and 2,348 in 2017**. In 2018, canceled orders dropped significantly to **86 orders**.
- **Shipping On Time:** On-time deliveries showed stable performance between 2015 and 2017, recording **10,985, 11,281, and 9,556 orders** respectively. However, 2018 reported only **374 on-time deliveries**.

Late Delivery Status Month Wise:

- **January:** January recorded the highest delivery activity, with **9,827 late deliveries, 4,162 advance shipments, and 3,300 on-time deliveries**. This indicates strong order volume but also a high number of delayed shipments.
- **February to May:** From February to May, late deliveries consistently remained high, ranging between **7,926 and 8,796 orders**. Advance shipping and on-time deliveries also stayed relatively stable, showing balanced operational performance during these months.
- **June to September:** During June to September, late deliveries continued to dominate, with August recording **8,878 delayed orders**, one of the highest monthly delay counts. Advance shipping and on-time delivery numbers remained steady throughout this period.
- **October to December:** The final quarter showed a decline in overall delivery activity. December recorded the lowest counts across all delivery categories, including **7,063 late deliveries and 2,252 on-time deliveries**, indicating reduced seasonal order volume.
- **Shipping Cancellations:** Shipping cancellations remained comparatively low throughout the year, fluctuating between **507 and 810 orders** monthly. May recorded the highest number of canceled shipments with **810 orders**.



Monthly, Weekday, Quarter and Yearly Trend

Yealy Revenue:

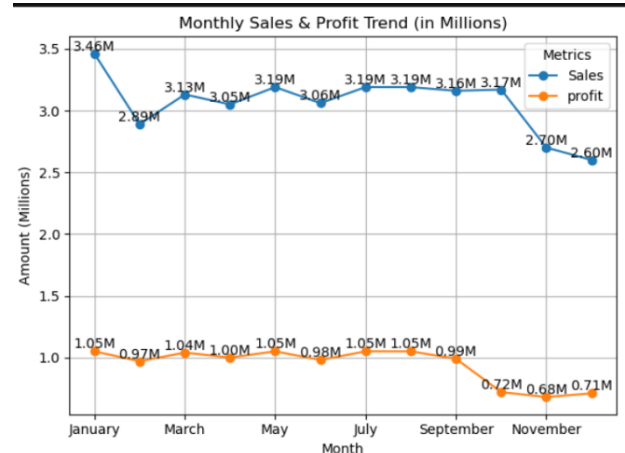
- **Year 2015:** In 2015, the company processed **62,650 orders**, generating total revenue of **\$12.34 Million** and profit of **\$1.32 Million**. This year recorded the highest overall business performance during the analysis period.
- **Year 2016:** The business handled **62,550 orders** in 2016, with total revenue of **\$12.30 Million** and profit of **\$1.31 Million**. Performance remained highly stable compared to 2015, indicating consistent market demand and operational efficiency.
- **Year 2017:** In 2017, total orders declined to **53,196**, while revenue reached **\$11.81 Million** and profit stood at **\$1.30 Million**. Although orders decreased, profitability remained relatively stable.
- **Year 2018:** The company recorded only **2,123 orders** in 2018, generating revenue of **\$0.33 Million** and profit of **\$0.03 Million**. The significant decline suggests incomplete yearly data or reduced operational activity.

Yearly Trend			
Year	Orders	Total Revenue	Total Profit
2015	62650	\$12.34 M	\$1.32 M
2016	62550	\$12.30 M	\$1.31 M
2017	53196	\$11.81 M	\$1.30 M
2018	2123	\$0.33 M	\$0.03 M
Grand Total	180519	\$36.78 M	\$3.97 M

Monthly Trend:

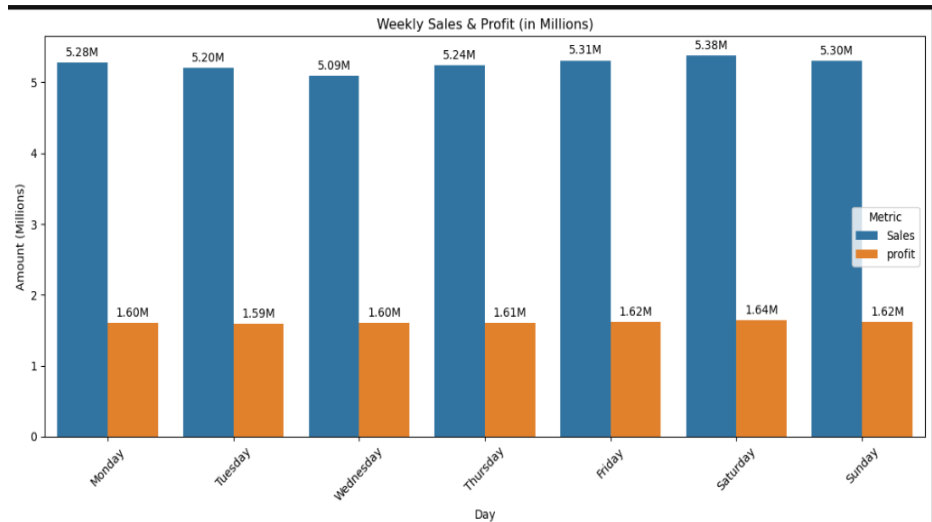
- **January:** January recorded the highest monthly performance with **17,979 orders**, generating **\$3.46 Million** in revenue and **\$0.37 Million** in profit. This indicates strong customer demand and high sales activity at the beginning of the year.
- **February:** In February, the company processed **14,529 orders**, earning **\$2.89 Million** in revenue and **\$0.30 Million** in profit. Sales performance slightly declined compared to January.
- **March:** March showed improved business performance with **15,919 orders**, generating **\$3.13 Million** in revenue and **\$0.33 Million** in profit, reflecting stable market demand.
- **April to September:** From April to September, monthly revenue remained relatively stable between **\$3.05 Million and \$3.19 Million**, while profits consistently stayed around **\$0.32–\$0.36 Million**. This period reflects balanced sales growth and operational stability.
- **October to December:** During the final quarter, orders and revenue gradually declined. December recorded the lowest monthly revenue of **\$2.60 Million** and profit of **\$0.28 Million**, indicating reduced customer activity or seasonal fluctuations.

Month_name	orders	Revenue_Mn	Total_Profit_Mn
January	17979	3.46	0.37
February	14529	2.89	0.3
March	15919	3.13	0.33
April	15435	3.05	0.34
May	15976	3.19	0.34
June	15139	3.06	0.32
July	15922	3.19	0.35
August	15912	3.19	0.36
September	15489	3.16	0.36
October	12955	3.17	0.33
November	12500	2.7	0.28
December	12764	2.6	0.28



Weekday Trend:

- **Friday:** Friday generated total sales of **\$5.31 Million** and profit of **\$1.62 Million**, reflecting strong customer purchasing activity toward the end of the workweek.
- **Monday:** On Monday, the company recorded **\$5.28 Million** in sales and **\$1.60 Million** in profit, indicating stable business performance at the beginning of the week.
- **Saturday:** Saturday achieved the highest sales and profit performance with **\$5.38 Million** in sales and **\$1.64 Million** in profit. This suggests increased customer shopping activity during weekends.
- **Sunday:** Sunday contributed **\$5.30 Million** in sales and **\$1.62 Million** in profit, maintaining strong weekend business performance.
- **Tuesday to Thursday:** Sales and profit remained relatively stable between Tuesday and Thursday, with daily sales ranging from **\$5.09 Million to \$5.24 Million** and profits around **\$1.59–\$1.61 Million**.



Quarterly Trend :

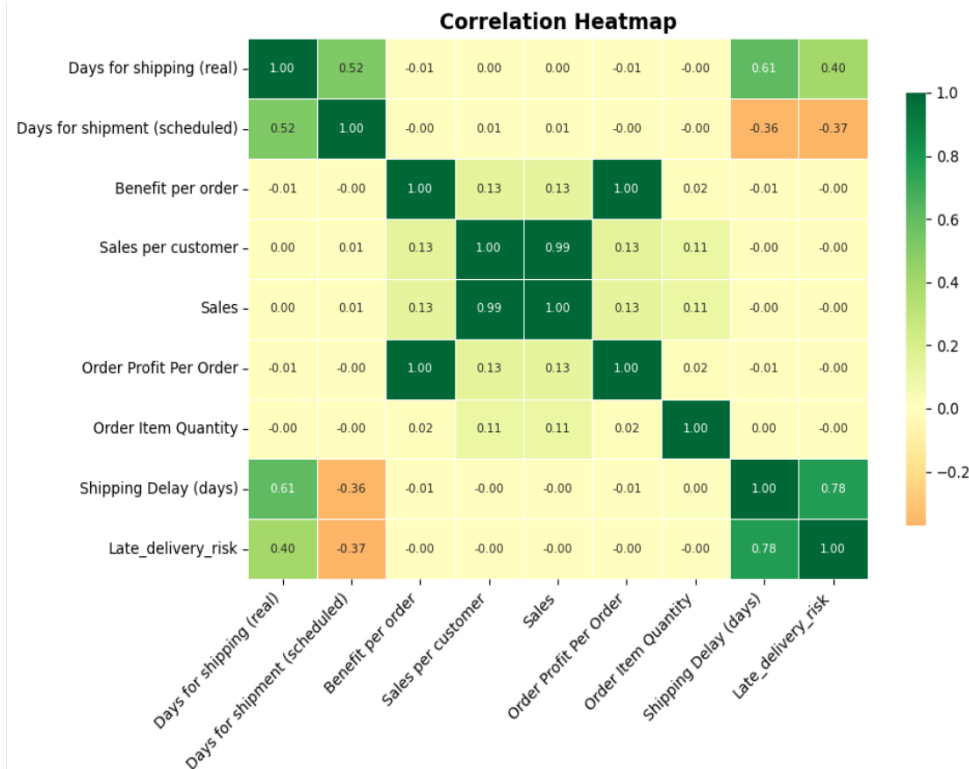
- **Year 2015:** In 2015, the company processed **62,650 orders**, generating total revenue of **\$12.34 Million** and profit of **\$1.32 Million**. Among all quarters, **Q4** recorded the highest revenue of **\$3.14 Million**, while **Q3** achieved the highest profit of **\$347,071.77**.
- **Year 2016:** The business handled **62,550 orders** in 2016, with revenue of **\$12.30 Million** and profit of **\$1.31 Million**. **Q4** generated the highest revenue at **\$3.13 Million**, whereas **Q3** delivered the highest quarterly profit of **\$354,055.97**.
- **Year 2017:** In 2017, total orders declined to **53,196**, while revenue reached **\$11.81 Million** and profit stood at **\$1.30 Million**. **Q3** became the strongest quarter with the highest revenue of **\$3.36 Million** and profit of **\$366,990.25**.
- **Year 2018:** The company recorded only **2,123 orders** in 2018, generating revenue of **\$0.33 Million** and profit of **\$33,841.89**. The lower values suggest incomplete yearly data or reduced business operations.
- **Overall Insight:** The quarterly trend analysis shows that business performance remained strong and stable from 2015 to 2017, with quarterly revenue consistently above **\$3 Million**. Most years

Quarterly Trend			
Year& Quarter	Orders	Revenue	Profit
2015	62650	\$12.34 M	\$1.32 M
Q1	15413	3029853.67	324579.61
Q2	15617	3088947.89	330378.74
Q3	15712	3085914.48	347071.77
Q4	15908	3136115.39	316826.78
2016	62550	\$12.30 M	\$1.31 M
Q1	15421	3040704.22	294850.43
Q2	15453	3033671.33	329888.81
Q3	15799	3096312.91	354055.97
Q4	15877	3133128.86	331323.86
2017	53196	\$11.81 M	\$1.30 M
Q1	15470	3070237.7	348643.82
Q2	15480	3175893.43	341374.99
Q3	15812	3357485.64	366990.25
Q4	6434	2204819.37	247076.05
2018	2123	\$0.33 M	\$0.03 M
Q1	2123	331650.12	33841.89
Grand Total	180519	36784735.01	3966902.974

achieved peak performance during **Q3 or Q4**, indicating stronger sales activity in the second half of the year. Overall, the company generated **180,519 orders**, **\$36.78 Million** in revenue, and **\$3.97 Million** in total profit.

Correlation Analysis Insights

- Days for Shipping (Real):** The heatmap shows that **Days for Shipping (real)** has a positive correlation of **0.61** with **Shipping Delay (days)** and **0.40** with **Late Delivery Risk**. This indicates that longer actual shipping times increase delivery delays and raise the chances of late deliveries.
 - Days for Shipment (Scheduled):** **Days for Shipment (scheduled)** has a moderate positive relationship of **0.52** with real shipping days. However, it shows a negative correlation of **-0.36** with shipping delay and **-0.37** with late delivery risk, suggesting that effective shipment scheduling can reduce delays and improve delivery performance.
- | | Days for shipping (real) | Days for shipment (scheduled) | Benefit per order | Shipping delay (days) | Late delivery risk | On-time delivery rate | Customer satisfaction | Logistics cost | Delivery time (days) |
|-------------------------------|--------------------------|-------------------------------|-------------------|-----------------------|--------------------|-----------------------|-----------------------|----------------|----------------------|
| Days for shipping (real) | 1.00 | 0.52 | -0.01 | 0.00 | 0.00 | -0.01 | -0.00 | 0.61 | 0.40 |
| Days for shipment (scheduled) | 0.52 | 1.00 | -0.00 | 0.01 | 0.01 | -0.00 | -0.00 | -0.36 | -0.37 |
| Benefit per order | -0.01 | -0.00 | 1.00 | 0.13 | 0.13 | 1.00 | 0.02 | -0.01 | -0.00 |



- Benefit per Order & Profitability:** The variables **Benefit per Order** and **Order Profit Per Order** have a perfect positive correlation of **1.00**, meaning both metrics are directly connected and strongly represent business profitability.
- Sales & Sales per Customer:** The strongest relationship in the heatmap is between **Sales** and **Sales per Customer** with a correlation value of **0.99**. This shows that increased customer spending has a direct positive impact on total sales revenue.
- Order Item Quantity:** **Order Item Quantity** has only weak correlations with sales and profitability variables, indicating that increasing product quantity alone does not significantly influence revenue or profit growth.
- Shipping Delay & Late Delivery Risk:** The correlation between **Shipping Delay (days)** and **Late Delivery Risk** is **0.78**, showing a strong positive relationship. This means that higher shipping delays greatly increase the probability of late deliveries and customer dissatisfaction.
- Overall Insight:** The correlation analysis highlights that shipping efficiency, transportation management, and customer spending behavior are the most important factors affecting supply chain performance. Improving shipment scheduling and reducing shipping delays can significantly enhance delivery performance, customer satisfaction, and overall operational efficiency.

Statistical Test:

TEST 1: ONE-SAMPLE T-TEST

- Is average shipping delay significantly different from 0?

One-Sample T-Test Analysis

- The One-Sample T-Test was conducted to determine whether the average shipping delay is significantly different from **0 days**. The analysis produced a **sample mean delay of 0.5658 days**, indicating that shipments are delayed by more than half a day on average.
- The test generated a very high **T-Statistic value of 161.2361** with a **p-value of 0.000000**, which is significantly lower than the standard significance level of 0.05. As a result, the **Null Hypothesis is rejected**.
- This statistical result confirms that the average shipping delay is not equal to zero and that delivery delays are statistically significant within the supply chain process. The findings indicate consistent shipping inefficiencies that may affect customer satisfaction and operational performance.
- Overall, the test highlights the need for improvements in logistics management, shipment planning, transportation efficiency, and warehouse operations to reduce delivery delays and improve overall supply chain performance.

```
Sample Mean = 0.5658 days  
T-Statistic = 161.2361 | p-value = 0.000000  
Reject Null Hypothesis  
→ Mean delay (0.57 days) is significantly different from 0.
```

Test 2. INDEPENDENT T-TEST

- Do late deliveries have significantly higher shipping delay?
- The Independent T-Test was performed to determine whether late deliveries have significantly higher shipping delays compared to on-time or advance deliveries. The analysis shows that **Late Deliveries** had an average shipping delay of **1.618 days** with a standard deviation of **0.883** across **98,977 orders**.
- In comparison, **On-Time and Advance Deliveries** recorded an average delay of **-0.712 days** with a standard deviation of **1.000** across **81,542 orders**, indicating that these orders were generally delivered earlier than the expected delivery date.
- The test produced a very high **T-Statistic value of 519.2561** and a **p-value of 0.000000**, which is significantly lower than the standard significance level of 0.05. Therefore, the **Null Hypothesis (H_0) is rejected**.
- This result confirms that there is a statistically significant difference in shipping delays between late deliveries and on-time/advance deliveries. The findings clearly indicate that delayed orders experience substantially higher shipping delays, highlighting operational inefficiencies in logistics and transportation management.

```
Late Delivery    - Mean: 1.618 | Std: 0.883 | n=98,977  
On-Time/Advance - Mean: -0.712 | Std: 1.000 | n=81,542  
T-Statistic = 519.2561 | p-value = 0.000000  
Statistically Significant (reject  $H_0$ )
```

Test -3. ONE-WAY ANOVA

Do shipping delays differ significantly across shipping modes?

- The One-Way ANOVA test was conducted to determine whether shipping delays differ significantly across different shipping modes. The analysis included **4**

```
Groups tested: 4 shipping modes  
Group means:  
First Class      → Mean: 1.000 days | n=27,814  
Same Day         → Mean: 0.478 days | n=9,737  
Second Class     → Mean: 1.991 days | n=35,216  
Standard Class   → Mean: -0.004 days | n=107,752  
  
F-Statistic = 23246.0527 | p-value = 0.000000  
Statistically Significant (reject  $H_0$ )
```

shipping modes: First Class, Same Day, Second Class, and Standard Class.

- The results show that **Second Class Shipping** recorded the highest average shipping delay of **1.991 days** across **35,216 orders**, indicating the slowest delivery performance among all shipping methods. **First Class Shipping** had an average delay of **1.000 day** with **27,814 orders**, while **Same Day Shipping** showed a much lower average delay of **0.478 days** across **9,737 orders**.
- In contrast, **Standard Class Shipping** recorded an average delay of **-0.004 days** with **107,752 orders**, indicating that most deliveries were completed close to or before the expected delivery date.
- The ANOVA test generated an extremely high **F-Statistic value of 23,246.0527** with a **p-value of 0.000000**, which is far below the significance level of 0.05. Therefore, the **Null Hypothesis (H₀)** is rejected.
- This statistical result confirms that shipping delays vary significantly across different shipping modes. The findings suggest that shipping method selection has a major impact on delivery performance and customer experience. Improving operational efficiency in slower shipping modes, especially **Second Class Shipping**, could help reduce delays and improve overall supply chain performance.

Test-4. CHI-SQUARE TEST

Is there a relationship between Shipping Mode & Delivery Status?

- The Chi-Square Test was conducted to determine whether there is a significant relationship between **Shipping Mode** and **Delivery Status**. The contingency table analysis included four shipping modes: **First Class**, **Same Day**, **Second Class**, and **Standard Class**.

Contingency Table (counts):				
Delivery Status	Advance shipping	Late delivery	Shipping canceled	Shipping on time
Shipping Mode				
First Class	0	26513	1301	0
Same Day	0	4454	444	4839
Second Class	0	26987	1410	6819
Standard Class	41592	41023	4599	20538

Chi2 Statistic = 55782.4835 | df = 9 | p-value = 0.000000
Statistically Significant (reject H₀)
→ Shipping Mode significantly affects Delivery Status.

- The results show that **First Class Shipping** recorded **26,513 late deliveries** and **1,301 canceled shipments**, with no advance or on-time deliveries observed. **Same Day Shipping** performed better, with **4,839 on-time deliveries** and comparatively fewer late deliveries (**4,454 orders**).
- **Second Class Shipping** experienced a very high number of delayed shipments, with **26,987 late deliveries** and **1,410 canceled orders**, indicating weaker delivery performance. In contrast, **Standard Class Shipping** handled the largest order volume and recorded **41,592 advance shipments** and **20,538 on-time deliveries**, showing relatively better operational efficiency.
- The Chi-Square test produced a very high **Chi2 Statistic value of 55,782.4835** with **9 degrees of freedom** and a **p-value of 0.000000**, which is significantly lower than the standard significance level of 0.05. Therefore, the **Null Hypothesis (H₀)** is rejected.
- This statistical result confirms that there is a strong and significant relationship between shipping mode and delivery status. The findings indicate that the type of shipping method directly influences delivery performance, late delivery risk, and customer service outcomes. Improving inefficient shipping modes, especially **Second Class** and **First Class Shipping**, could help reduce delays and improve overall supply chain efficiency.

Test-5 CHI-SQUARE TEST

- **Is there a relationship between Customer Segment & Late Delivery Risk?**
- The Chi-Square Test was conducted to determine whether there is a significant relationship between **Customer**

Contingency Table:		
Late_delivery_risk	0	1
Customer Segment		
Consumer	42256	51248
Corporate	24807	29982
Home Office	14479	17747

Chi2 = 1.0247 | df = 2 | p-value = 0.599074
Not Statistically Significant (fail to reject H₀)

Segment and Late Delivery Risk. The analysis included three customer segments: **Consumer, Corporate, and Home Office**. The contingency table shows that the **Consumer segment** recorded **51,248 late delivery risk orders** and **42,256 non-risk orders**, making it the largest customer segment in the dataset. The **Corporate segment** had **29,982 late delivery risk orders** and **24,807 non-risk orders**, while the **Home Office segment** recorded **17,747 late delivery risk orders** and **14,479 non-risk orders**.

- The Chi-Square test produced a **Chi2 Statistic value of 1.0247** with **2 degrees of freedom** and a **p-value of 0.599074**, which is significantly higher than the standard significance level of 0.05. Therefore, the **Null Hypothesis (H_0) cannot be rejected**.
- This statistical result indicates that there is **no significant relationship** between customer segment and late delivery risk. In other words, the probability of late delivery remains relatively similar across Consumer, Corporate, and Home Office customers.

Test-6. PEARSON CORRELATION

Is there a linear relationship between Discount Rate & Profit Ratio?

- The Pearson Correlation Test was conducted to examine whether there is a linear relationship between **Discount Rate** and **Profit Ratio**. The analysis produced a **Pearson correlation coefficient (r) of -0.0027**, indicating an extremely weak negative relationship between the two variables. The test also generated a **p-value of 0.252934**, which is greater than the standard significance level of 0.05. Therefore, the **Null Hypothesis (H_0) cannot be rejected**, meaning the relationship is **not statistically significant**.
- This result suggests that changes in discount rates do not have a strong or consistent impact on the profit ratio within the dataset. Although the correlation is slightly negative, the relationship is too weak to conclude that higher discounts significantly reduce profitability.

Pearson r = -0.0027 | p-value = 0.252934
Not Statistically Significant (fail to reject H_0)
→ Weak negative correlation between discount and profit ratio.

Test-7. SPEARMAN CORRELATION (Non-parametric)

Relationship between Real Shipping Days and Shipping Delay

- The Spearman Correlation Test was conducted to examine the relationship between **Real Shipping Days** and **Shipping Delay** using a non-parametric statistical approach. The analysis produced a **Spearman correlation coefficient (ρ) of 0.6040**, indicating a moderately strong positive relationship between the two variables. The test generated a **p-value of 0.000000**, which is significantly lower than the standard significance level of 0.05. Therefore, the **Null Hypothesis (H_0) is rejected**, confirming that the relationship is statistically significant.
- This result indicates that as the number of real shipping days increases, shipping delays also tend to increase. In other words, longer transportation and delivery durations are strongly associated with delayed shipments.
- The positive correlation highlights the importance of improving transportation efficiency, route optimization, warehouse coordination, and logistics planning to minimize delivery delays and improve overall supply chain performance.

Spearman ρ (rho) = 0.6040 | p-value = 0.000000
Statistically Significant (reject H_0)

Conclusion:

The analysis of 180,519 global orders revealed strong overall business performance, with total revenue of **\$36.78 Million** and total profit of **\$3.97 Million** generated across multiple markets, regions, and product categories. Europe and LATAM emerged as the top-performing markets, while categories such as Fishing, Cleats, Camping & Hiking, and Cardio Equipment contributed the highest profitability.

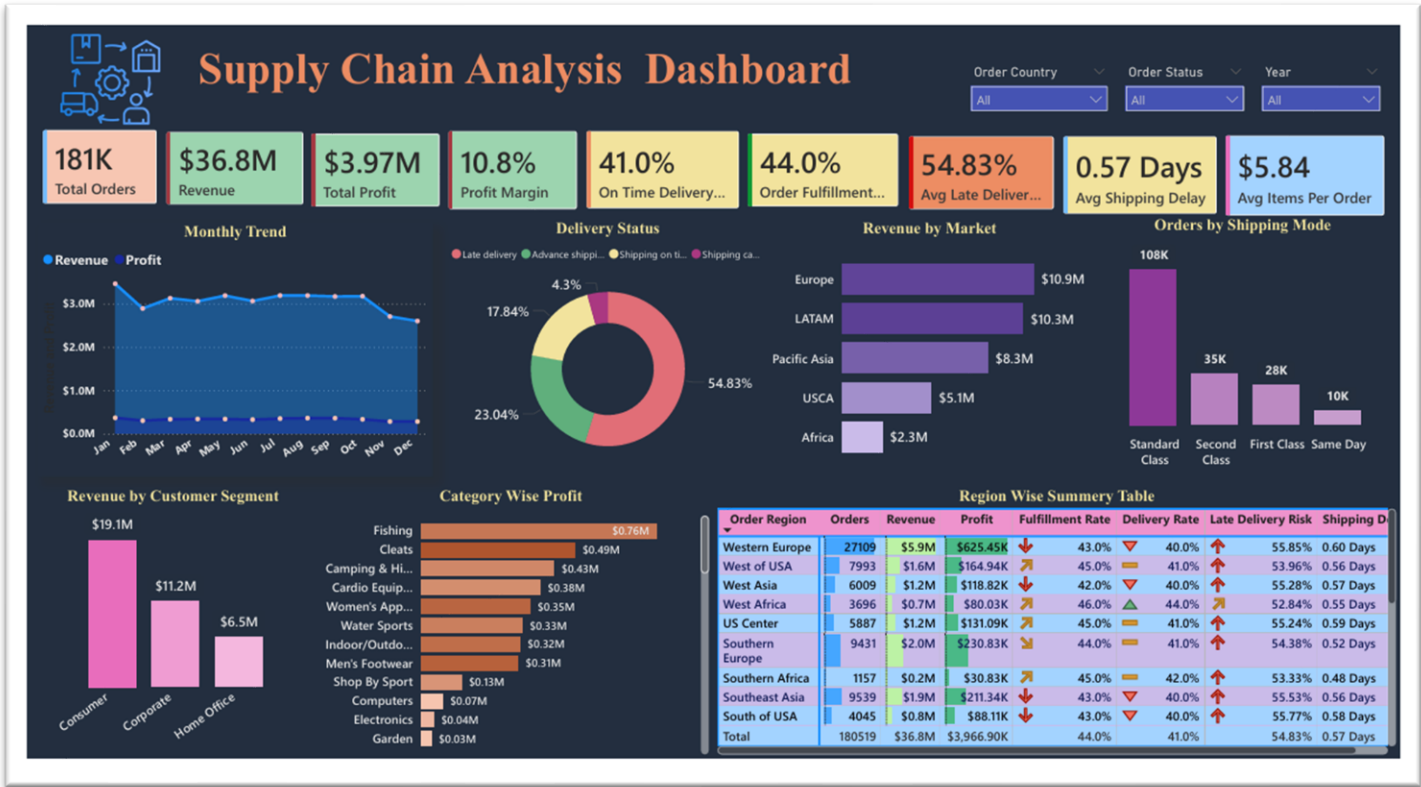
Despite strong sales performance, the project identified several critical operational and logistics challenges affecting supply chain efficiency. The company recorded a high **Late Delivery Rate of 54.83%**, and more than half of all orders experienced delivery delays. Statistical tests including the One-Sample T-Test, Independent T-Test, ANOVA, Chi-Square Test, and Spearman Correlation confirmed that shipping delays are statistically significant and strongly influenced by shipping mode selection and real shipping duration.

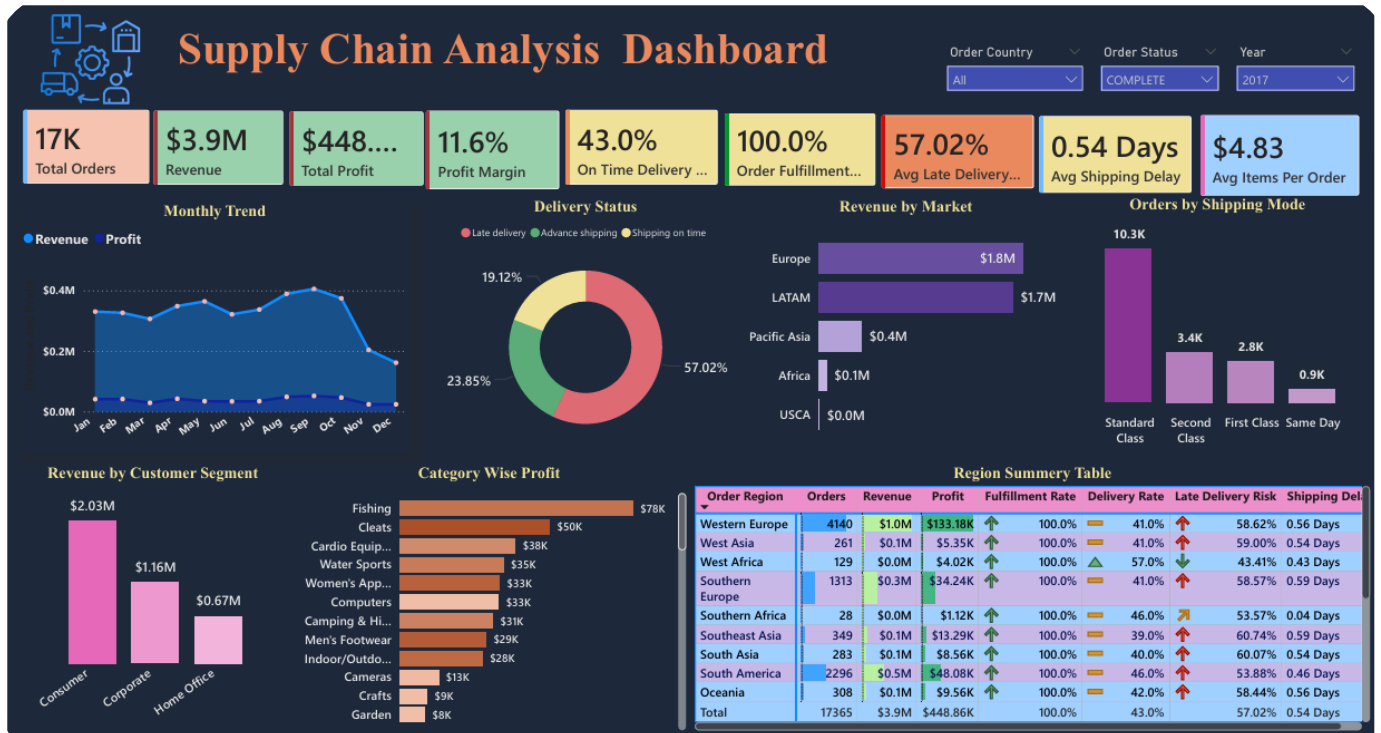
The analysis also revealed major profitability concerns. Nearly **18.71% of total orders were loss-making**, while several products and categories generated very low revenue and weak margins. Outlier analysis further highlighted abnormal discounting practices and extreme negative profit values, indicating revenue leakage and inefficient pricing strategies.

Customer behavior analysis showed that the Consumer segment contributed the highest sales revenue, while weekend sales activity — especially on Saturdays and Sundays — generated the strongest business performance. Product analysis demonstrated that sportswear, outdoor recreation, and fitness-related products dominate customer demand and revenue generation.

Correlation analysis emphasized that shipping efficiency and transportation management are the most important operational factors affecting customer satisfaction and delivery performance. The strong relationship between shipping delays and late delivery risk highlights the urgent need for logistics optimization and better shipment planning.

Power Bi Dashboard





Recommendations:

1. Improve Logistics & Delivery Operations

The company should focus on reducing the high late delivery rate by improving transportation planning, warehouse coordination, and route optimization. Special attention should be given to Second Class Shipping, which recorded the highest average delivery delay.

2. Optimize Shipping Mode Strategy

Standard Class and Same Day Shipping showed better delivery performance compared to First Class and Second Class Shipping. The business should encourage the use of more efficient shipping modes and redesign inefficient delivery channels.

3. Reduce Loss-Making Orders

Approximately 18.71% of orders generated financial losses. The company should review pricing strategies, shipping costs, discount policies, and low-margin products to minimize revenue leakage and improve profitability.

4. Control Discounting Practices

Outlier analysis revealed abnormal discounts reaching up to \$500 on some orders. The organization should implement stricter discount approval policies and monitor promotional campaigns to protect profit margins.

5. Focus on High-Performing Categories

Categories such as Fishing, Cleats, Camping & Hiking, and Cardio Equipment generated the highest profits. Increasing inventory availability, targeted promotions, and marketing investments in these categories can further boost revenue growth.

6. Improve Low-Performing Categories

Low-revenue categories such as CDs, Toys, Books, and Golf Bags & Carts should be evaluated for product demand, pricing effectiveness, and customer interest. Underperforming products may require repositioning or discontinuation.

7. Strengthen Market Expansion Strategies

Europe and LATAM contributed the highest revenue and profit. The company should continue expanding operations in these high-performing markets while developing targeted growth strategies for lower-performing regions such as Africa and USCA.

8. Enhance Payment & Order Processing Systems

A large number of orders remained under Pending Payment and Processing status. Improving payment verification systems, customer communication, and order tracking can reduce operational delays and improve customer satisfaction.

9. Use Predictive Analytics for Demand Forecasting

The organization should adopt predictive analytics and machine learning models to forecast customer demand, optimize inventory levels, and improve supply chain planning accuracy.

10. Build Real-Time Supply Chain Dashboards

Implementing interactive real-time dashboards in Power BI can help management continuously monitor KPIs such as late delivery risk, shipping performance, profitability, inventory status, and customer demand patterns for faster decision-making.



Thankyou And Happy Learning

Author's Name- Bipin Chandra Arya